

A Robot Is What It Becomes: Toward Ecological Robotics

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The question “What is a robot?” does not have a single answer. It depends on who is asked—and when. Engineers may define a robot through sensing, actuation, and autonomy. HCI researchers may describe it as a medium for interaction. Industry may treat it as a scalable technological product. These definitions also shift across a system’s lifecycle: during development, laboratory study, deployment, and long-term integration. The definition we adopt is not neutral. It shapes how robots are envisioned, evaluated, and ultimately designed. When robots are understood primarily as technical artifacts, design emphasizes hardware optimization and efficiency. When framed as interaction media, emphasis shifts toward usability and user experience. Yet what ultimately emerges from these processes is not only a device, but a persistent presence embedded in complex real-world environments—environments that extend across time and involve actors beyond the human user alone. We propose *Ecological Robotics* as a shift in perspective within Human–Robot Interaction. Instead of designing robots bottom-up—from hardware to interaction to application—we suggest beginning with the ecological role a robot is intended to assume in the world. From this standpoint, technical architecture, interaction design, and material choices follow from a prior commitment to what the robot should become once deployed.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; **HCI theory, concepts and models**.

Additional Key Words and Phrases: Human-Robot Interaction, Ecological Robotics

1 INTRODUCTION

Robotics has long resisted a stable definition. As Bernard Roth noted in the foreword to the Springer Handbook of Robotics [11], the question of what counts as a robot has “never been successfully resolved” and may remain a moving target as technology evolves. Early robotics centered on mechatronic devices combining sensing, actuation, and control. Later developments incorporated perception, planning, and artificial intelligence, leading to broader definitions of robots as machines capable of intelligently relating perception to action in the physical world. Over time, robotics expanded further to include human-centered and social dimensions [3, 9]. This shift reflects a growing recognition that robots are not only technical artifacts, but entities operating within social and lived environments.

At this stage of technological development, one may question whether proposing yet another abstract definition of “robot” is useful at all. As technologies evolve rapidly and continuously blur boundaries, a certain degree of ambiguity—paired with a shared but flexible understanding—may be more productive than a rigid definition. Attempting to carve a fixed meaning “in stone” risks partitioning fields, devices, and research trajectories that would benefit from being examined together. Overly strict definitions may exclude emerging forms of embodied AI or distributed systems that challenge traditional categories. Rather than seeking a final and stable definition, we suggest shifting the focus toward the consequences of how robots are positioned in the world. The question is not only what a robot is in abstract terms, but what changes when we define it in one way rather than another. If definitions evolve as robots enter new domains of life, then perhaps what defines a robot is not a fixed set of internal properties, but its situated mode of existence. Echoing Jean-Paul Sartre’s statement that “existence precedes essence” [10], we argue that a robot’s defining

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characteristics may emerge less from its internal design and more from the position it comes to occupy within a system of relations. A robot is not only what it is built to be; it is what it becomes in the world.

To frame this shift, we introduce the notion of ecological robotics. By this, we refer to the relational position a robot comes to occupy within an ecosystem (i.e., a concrete configuration of relations within a bounded environment), shaping and being shaped by surrounding actors and material conditions. While socio-technical systems perspectives often emphasize structural interdependence between social and technical elements, an ecological perspective focuses more specifically on role formation, niche occupation, and long-term stabilization. The emphasis shifts from how components are connected to how a robot comes to be recognized, through its design and mode of existence, as a particular kind of inhabitant within a system.

2 DESIGNING FOR ECOLOGICAL ROBOTS

If we accept this premise, design changes fundamentally. Instead of starting with hardware, then defining interaction modalities, and finally identifying applications, we begin by asking: **Who or what do we want this robot to become within its target ecosystem?**

This perspective connects closely to the use of metaphors in HCI [1, 8] and HRI [6]. Labels such as “assistant,” “co-pilot,” “companion,” or “pet” are not neutral descriptions. They project anticipated roles and prefigure patterns of coexistence. They shape how the robot is expected to behave, how it should be treated, and how it will be recognized within an ecosystem. As a result, even mechanically identical systems can become ontologically distinct depending on the ecological role they are designed to stabilize.

Designing for ecological role also encourages a broader understanding of sustainability—both environmental and social. When robots are conceived primarily as technical products, attention often centers on performance, efficiency, and rapid iteration. In contrast, designing from the standpoint of ecological integration requires considering long-term stabilization within an ecosystem. This includes material aspects such as composition, durability, maintenance, repairability, and end-of-life consequences, but extends further. A robot’s existence has territorial implications: how it occupies space, how it moves through shared environments, whether it yields or claims priority, and how it reshapes proxemic norms. Its sensory presence, such as sound, light, smell, or data collection, also affects how space is experienced by other stakeholders. Even subtle traces may alter patterns of coexistence. For each of these dimensions, an ecological robot should be designed in alignment with its intended role. This perspective may challenge implicit assumptions in HRI about how robots ought to behave. For example, a robot intended to function as a recognized member of a family might need to exhibit forms of territoriality, such as maintaining personal space or signaling boundaries, rather than always yielding passively. Ecological robots, in this sense, are not necessarily designed to be frictionless or universally accommodating, but to convincingly enact the roles they are meant to assume within a given ecosystem. In this view, interaction design does not disappear; rather, it becomes one expression of ecological role—shaped by the kind of inhabitant the robot is intended to be.

Importantly, an ecological role does not necessarily imply a deep social relationship. A role can also be that of a tool, infrastructure element, or background agent. Popular culture offers a useful illustration. In Star Wars, robotic entities occupy a wide spectrum of ecological roles. C-3PO functions as a diplomatic mediator; R2-D2 and BB-8 act as collaborative co-pilots; GNK (“Gonk”) droids and mouse droids serve infrastructural functions; B1 battle droids operate as disposable military agents. Although all are considered robots, their social positioning, autonomy, and legitimacy differ dramatically. What distinguishes them is not simply their technical capability, but the niche they occupy within their ecosystem.

A more grounded speculative example further illustrates this shift. Imagine walking in a forest and hearing rustling among the leaves. You notice a small autonomous FoRest Exploration Device (FRED), designed to conduct environmental analysis while blending seamlessly into wildlife ecology (see Fig. 1). It freezes when observed, avoids direct human approach, and moves in ways reminiscent of an animal. FRED is not optimized for human engagement, but for ecological legitimacy. Its role is not that of a visible forest guard or overt scientific instrument, but closer to that of a wildlife-like inhabitant. Extending this logic further, FRED could draw energy from forest biomass and, when it ceases functioning, decompose naturally through biodegradable materials. Such choices align its material existence with its ecological role. Behavioral, material, and energetic design decisions thus work together to preserve the atmosphere of wildness while minimizing disruption to other actors in the ecosystem.



Fig. 1. A small autonomous robot collecting ecological data in the rainforest, illustrating what could be a prototype of FRED (Photo: Claus Melvad, in [5]).

3 ECOLOGICAL ROLE

What ecological role should a robot assume? There is no single answer. Social robotics has already explored broad roles such as companion [2], assistant [7], or emotional support agent [4], even if not framed explicitly in ecological terms. Ecological robotics invites a more systematic reflection: what mode of ecological existence should a robot adopt once deployed within a given ecosystem? Here, the granularity of the role becomes crucial. A role can be defined broadly or narrowly. The broader the role, the wider the associated design space; the more specific the role, the more constrained and directed the design becomes. Popular culture illustrates this dynamic. In *Detroit: Become Human*, technically similar humanoid robots, designed to resemble humans, are marketed as caregivers, companions, or coaches. Although their technical architectures are comparable and they share human-like embodiment, each assigned role shapes their appearance, behavior, and social positioning differently. The distinction lies less in hardware than in the ecological niche each robot is intended to occupy.

Importantly, if existence precedes essence, ecological role is not determined solely by design intention. A role refers to the intended relational script projected through a robot's design, how it is meant to behave, to be perceived, and to position itself within an ecosystem. However, the stabilization of that role into an ecological niche, that is, the enacted and sustained configuration of relations through which the robot's presence becomes embedded within the ecosystem, depends on how other actors respond in practice. While a robot may clearly express its intended role through its mode of existence, friction can arise if surrounding actors do not engage in accordance with the expected relational dynamic. For instance, if people attempt to catch FRED and play with it rather than treat it as one would a wild animal, the intended positioning fails to stabilize. Ecological positioning is therefore relational and co-constructed rather than imposed. This dynamic is not unique to robots but reflects broader processes within ecosystems. For example, a person whose understanding of wildlife is shaped by fictional portrayals might approach a wild animal as if it were harmless, misjudging the ecological relationship at play. In contrast, someone with ecological knowledge or lived experience would recognize appropriate relational boundaries. Similarly, a robot's intended role may be carefully designed and enacted, yet fail to stabilize into a niche if it is interpreted differently by those with whom it shares its environment.

4 EMBODIMENT AND ECOLOGICAL CONTINUITY

As artificial agents increasingly operate across distributed infrastructures and transferable embodiments, the assumption that a robot is a bounded physical system combining hardware and software within a single chassis becomes unstable. The question shifts from what a robot is made of to what persists when embodiment changes.

Consider a humanoid domestic robot functioning as a roommate. When its user leaves home and humanoid robots are restricted from public streets, it transfers to the user’s mobile phone. Later, it occupies a rented aerial embodiment to accompany the user on a hike, before returning to its home chassis. Across these transitions, the user experiences continuity: the same conversational history, personality traits, relational expectations, and shared memories. From a hardware perspective, three distinct physical systems are involved. From a relational perspective, there is one enduring entity. This scenario challenges conventional definitions. If a robot is equated with its chassis, each embodiment constitutes a different robot. If it is equated with software, embodiment becomes incidental. Neither account explains what is actually preserved. What remains constant is the ecological role the system enacts—a stable relational position within a lived ecosystem. We call this persistence ecological continuity: the endurance of a robot’s role and niche across shifting material instantiations. The robot is recognized not by the body it occupies, but by the role it continues to perform—roommate, companion, co-inhabitant. Embodiment becomes a vehicle for participation rather than the essence of identity. This perspective shifts the locus of robot identity from material boundaries to sustained patterns of participation. In ecological terms, a robot is defined less by its chassis than by the niche it occupies within a network of actors. Embodiments may change; ecological positioning may endure. Such a view complicates traditional understandings of robots as discrete physical artifacts. If an ecological role migrates across bodies, what anchors identity? Conversely, if multiple embodiments enact divergent roles, do they remain one robot or become distinct ecological actors? By foregrounding ecological continuity, the question moves from “Where is the robot?” to “What role is being enacted?” The robot becomes less a thing and more a stabilized mode of participation within an ecosystem. Embodiment is contingent; ecological role is constitutive.

5 CONCLUSION

In this workshop contribution, we have proposed a different perspective on Human–Robot Interaction. Before focusing on interaction modalities, forms, or applications, we suggest stepping back to consider a more encompassing question: what ecological role is the robot intended to assume once deployed? From this standpoint, technical architecture and interaction design become secondary—important, but contingent upon a broader vision of what the robot should become in the world. By foregrounding ecological role, we aim to bring relational, social, societal, ethical, spatial, and environmental considerations closer to the core of design decisions rather than treating them as downstream concerns. This contribution doesn’t offer a fixed framework, but introduces a lens intended to spark discussion. We invite HRI researchers to reflect on their own contributions and to ask: “What do I want my robots to become?” In doing so, design decisions and research explorations can be reconsidered through an ecological lens.

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